

1. Classifier Exploration

Caltech

kNN ($k = 13$) from the starter code was the baseline. I added **Gaussian Naive Bayes (GNB)**, which models each feature as a bell curve per class and picks the more likely class. It only has to estimate a mean and variance per feature, which seemed reasonable with only about 19 images per person. I also added **RidgeClassifierCV**, a linear classifier with built-in regularization; it picks the penalty strength from $\{0.1, 1, 10, 100\}$ by internal cross-validation, which should help when there are far more features (4,624) than images.

Features	Classifier	LOO genuine acceptance	5-fold identification acc.	Verification balanced acc.	Verification FAR
Raw distances	kNN ($k = 13$)	0.371	0.600	0.650	0.001
Raw distances	GNB	0.776	0.581	0.830	0.084
Raw distances	RidgeCV	0.563	0.760	0.813	0.003
Normalized coords.	RidgeCV	0.203	0.819	0.575	0.000

Table 1. Caltech results (68 landmarks).

kNN was the weakest model. With $k = 13$, a query is only accepted if at least 7 of its 13 nearest neighbors belong to the same person, which rarely happens when each person has about 19 images out of 374; in verification it almost never accepted anyone (genuine acceptance 0.30). GNB had the best verification balance (0.83) and the best LOO score, but it got there by being lenient: its FAR of 8.4% is far higher than any other model. RidgeCV was the stronger identifier (0.76) and almost never accepted an impostor, at the cost of rejecting more genuine users. In short, GNB trades security for convenience, while Ridge does the opposite.

SoF

For SoF I tried two different classifiers. **Nearest Centroid** averages the genuine and impostor templates into one prototype each and assigns a query to the closer one. **SGDClassifier** trains a linear SVM (hinge loss) with stochastic gradient descent. I standardized the features first, used balanced class weights because genuine templates are only a small fraction of the training set, and set `max_iter = 5000` with `random_state = 42`.

Features	Classifier	LOO genuine acceptance	5-fold identification acc.	Verification balanced acc.	Verification FAR
Raw distances	Nearest Centroid	0.752	0.445	0.667	0.363
Raw distances	SGD	0.782	0.467	0.816	0.176
Distance ratios	SGD	0.794	0.564	0.788	0.242

Table 2. SoF results (68 landmarks).

SGD beat Nearest Centroid on every metric. The biggest difference was security: Nearest Centroid accepted 36% of impostors, compared with 18% for SGD. Nearest Centroid squeezes all 22 other people into one average “impostor” prototype, which sits near the middle of face space, close to many real faces. SGD instead learns a weight for every distance and uses balanced class weights, so it can draw a much sharper boundary. Both models identified people far less accurately than on Caltech (0.45–0.47 vs. 0.58–

0.76). SoF is a harder dataset: the photos include glasses and lighting changes that can shift the eye landmarks, and some people have only 2–4 images while others have more than 30.

2. Feature Representation

Caltech: normalized landmark coordinates

Instead of distances, I used the landmark positions themselves. For each face I subtracted the mean landmark position, divided by the distance of the farthest landmark from that center, and flattened the result into 136 values. The goal was to remove where the face sits in the image and how large it is, so the classifier only sees the face's shape. It also shrinks the feature vector from 4,624 to 136 values.

I tested this with RidgeCV. Identification improved from 0.76 to 0.82, the best Caltech result overall, so the normalized coordinates clearly carry identity information. Verification, however, got much worse (balanced accuracy $0.81 \rightarrow 0.58$), because Ridge almost stopped accepting genuine users (LOO $0.56 \rightarrow 0.20$). In identification Ridge only needs the right person to score highest among 20 people, but in verification it has to make a yes/no call where only about 5% of the training examples are genuine, and Ridge has no class weighting. With the new features its genuine scores rarely crossed the decision threshold. Adding `class_weight = "balanced"` would be my next step.

SoF: normalized distance ratios

For SoF I kept the pairwise distances but divided each one by a face-scale factor: the diagonal of the bounding box around the landmarks. This keeps all the relationships between landmarks while removing overall face size, which varies with the distance to the camera. Unlike the Caltech representation, it is built from distances, so it also does not depend on how the face is rotated in the image.

I tested this with SGD. Identification improved noticeably, from 0.47 to 0.56, the largest gain on SoF, and the LOO score rose slightly ($0.78 \rightarrow 0.79$). Verification balanced accuracy dropped a little ($0.82 \rightarrow 0.79$), because the FAR rose from 18% to 24% while genuine acceptance barely changed. Removing size made people easier to tell apart in identification, but faces of different people also look more alike without it, so more impostors fell on the genuine side of the yes/no boundary.

Conclusion

The classifier and the feature representation mattered about equally, and the answer depended on the metric. GNB and RidgeCV sat at opposite ends of the convenience–security trade-off, and SGD clearly beat Nearest Centroid. Both size-normalized representations followed the same pattern: better identification, worse verification. Measuring more than the starter metric exposed trade-offs the original score hid.